

LIGHT PASS

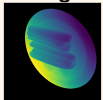
rasterize from the light $\mathbf{p}$

ϕ -MLP

3D Gaussians

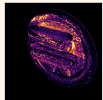
$\{\mu_i, \Sigma_i, o_i, \mathbf{b}_i, \mathbf{f}_i\}$

Light-space atlas



depth moments

M_1, M_2



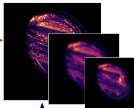
flux features

Φ

$$\mathbf{A}(u) = \sum_i w_i(u) [\phi_i, z_i, z_i^2, 1]$$

Pyramid

blur $\downarrow 2$ $\mathbf{A}_0, \dots, \mathbf{A}_{K-1}$



sample

CAMERA PASS

Receivers $\mathbf{x}$

Centers (training)
/ pixels (refinement)

project into light: $(u(\mathbf{x}), z(\mathbf{x}))$

Gather

$$\begin{aligned} \mathbf{s}_k &= [M_{0,k}, t_k, \delta_k, \log s_k, V_k], \quad \Phi_k \\ V_k &= 1 - M_{0,k} \Phi_{\mathcal{N}}(t_k - \beta) \end{aligned}$$

Learned shading heads

visibility g

$$V = \sigma(\text{logit } V_0 + g(\mathbf{S}, \mathbf{f}, \mathbf{n} \cdot \mathbf{l}))$$

transfer h

$$L_{\text{tr}} = \sum_k \mathbf{W}_k \Phi_k$$

material

$$\rho, s = \text{MLP}(\mathbf{f}, \mathbf{n}, \mathbf{l}, \mathbf{v}, \mathbf{h}, \mathbf{r})$$

$$L(\mathbf{x}) = E(\mathbf{x}) [V\rho + V_0 s + \sum_k \mathbf{W}_k \Phi_k], \quad E = I / \|\mathbf{p} - \mathbf{x}\|^2$$

render



I. Geometry warm-up (0–8k)

Point light; no networks;
target exponent $1/\gamma \rightarrow 1$

II. Joint training (8k–30k)

Linear radiance; shade Gaussians; densify

III. Refine appearance (+8k)

Frozen geometry; shade pixels
display-domain loss